

TTT4181 - Assignment 1

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1 Introduction

Sound-wave propagation in real-life environments is influenced by several factors, such as distance, obstacles and the acoustic properties of surrounding surfaces. Simulation software and numerical methods might be used to examine both free-field propagation, as well as obstacle bound ones. One such method is the Transmission Line Matrix (TLM), which is a time-domain numerical simulation technique developed for electromagnetism (and later adapted to acoustics). In this work, TLM is used to simulate and investigate the following phenomena within two spatial dimensions;

- propagation in free space
- reflection from a ground surface
- noise reduction from a screen

The scope of the experiments provide a brief introduction to the TLM method and demonstrate fundamental acoustic propagation phenomena.

2 Theory

This section presents the theoretical concepts required to understand and interpret the experiments described in this work. This includes the TLM method, sound pressure (level), free field propagation, reflection/interference and insertion loss.

2.1 TLM method

As mentioned in section 1, the Transmission Line Matrix (TLM) method is a numerical approach to solving wave problems in the time domain. Kagawa et al. provide a detailed treatment of the discrete Huygens' model approach to sound wave propagation using TLM [1].

2.1.1 From continuous physical problem to numerical model

In two spatial dimensions, the acoustic pressure field can be represented as $p(x, y, t)$, where x and y describe spatial position and t denotes time. Since the TLM method is implemented numerically on a computer, the physical system must be discretized in space and time, so that we might calculate the wave propagation numerically.

Kagawa describes Huygens' principle as a model in which a wavefront can be considered as a collection of secondary point sources that generate spherical wavelets [1]. This view enables us to consider wave propagation as a superposition of local contributions. Kagawa then expands on this idea through the

TLM method, where the physical model of the field in which the wave propagates is discretized through a network of nodes and transmission lines. Figure 1 illustrates this concept:

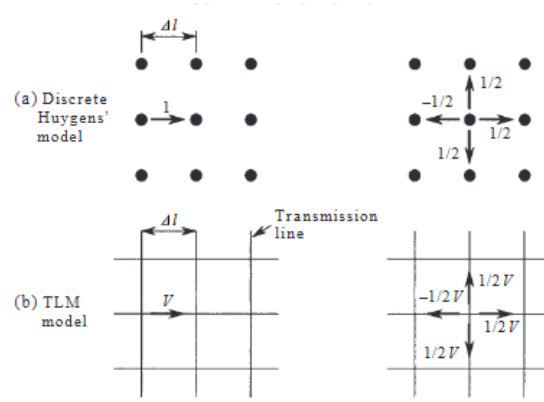


Figure 1: Illustration of the TLM concept through nodes and transmission lines, adapted from [1].

2.1.2 TLM nodes and scattering

We have established that the physical field is represented by a grid of nodes, connected by transmission lines. Waves approaching along the transmission lines are called *incident waves*. At the nodes, these are subject to a scattering process, in which the result is outgoing waves propagating towards the neighbouring nodes (see Figure 1). This process is repeated at discrete time steps. Ultimately, the TLM method applies this discrete representation in order to calculate numerically how the acoustic field evolves with respect to space and time [1].

The physical domain in TLM is represented by a discrete grid of cells. The spatial resolution of this grid is therefore related to the wavelength of the simulated sound. In TLMfig, the number of cells per wavelength (NCPW) specifies how many grid cells are used to represent one wavelength [2]. A higher number of cells per wavelength provides a finer spatial representation of the wave.

2.2 Sound pressure & sound pressure level

As stated in subsubsection 2.1.1, the acoustic pressure field in two-dimensional space can be represented by $p(x, y, t)$. Here, $p(x, y, t)$ represents the time-dependent acoustic pressure variation around the static atmospheric pressure. In order to describe the effective magnitude of the pressure variations, the RMS-value is introduced. RMS is calculated as seen in Equation 1.

$$p_{rms} = \sqrt{\frac{1}{T} \int_0^T p^2(t) dt} \quad (1)$$

The RMS value provides a measure of the effective magnitude of the pressure variation, independent from sign [3]. Since the TLM method assumes discretization, the formula becomes:

$$p_{rms} = \sqrt{\frac{1}{N} \sum_{n=1}^N p^2[n]} \quad (2)$$

The introduction of p_{rms} enables us to define *sound pressure level* (SPL) as follows:

$$L_p = 20 \log_{10} \left(\frac{p_{rms}}{p_0} \right) \quad (3)$$

where $p_0 = 20 \mu\text{Pa}$ is the reference sound pressure in air [3].

2.3 Free field propagation

A free field is an acoustic environment in which sound propagates without reflections from boundaries [3]. Thus, the sound pressure level can be examined as a function of distance from the sound source without contributions from boundary reflections.

As the distance from the source increases, the wavefront will expand over a greater area. Consequently, the amplitude of pressure must decrease. At this point, it is necessary to differentiate between idealized 3D and 2D cases.

For an idealized 3D point-source:

$$p(r) \propto \frac{1}{r} \quad (4)$$

Whilst an idealized 2D line-source yields:

$$p(r) \propto \frac{1}{\sqrt{r}} \quad (5)$$

[3]. Since TLMfig models sound propagation in two spatial dimensions, the 2D relationship is the relevant theoretical reference for the simulations in this work.

Using the definition of SPL from Equation 3, we can describe a difference of levels between two distances as follows:

$$L_p(r) - L_p(r_0) = 20 \log_{10} \left(\frac{p(r)}{p(r_0)} \right) \quad (6)$$

Inserting for the 2D-case yields:

$$L_p(r) - L_p(r_0) = -10 \log_{10} \left(\frac{r}{r_0} \right) \quad (7)$$

Thus, we should expect that a doubling in distance to reflect a change in SPL of:

$$\Delta L_p = -10 \log_{10}(2) \approx -3.0 \text{ dB} \quad (8)$$

This provides a theoretical expectation to which the result from the TLM simulations might be compared.

2.4 Reflection & interference

2.4.1 Reflection of a surface

When a sound wave encounters a surface, acoustic energy may be reflected, transmitted, or absorbed [3]. We will focus primarily on the reflected part. For an ideally hard surface, the reflection factor has unit magnitude, whereas an absorbing surface has a lower reflection factor.

Assuming plane waves normally incident to the surface, as well as the surface being locally reacting, the pressure reflection factor can be defined as the ratio between the magnitudes of the reflected and incident acoustic pressures [3]:

$$R = \frac{|p_{ref}|}{|p_{inc}|} \quad (9)$$

2.4.2 Interference due to reflection

As a consequence of reflection, a microphone in the field may receive both the direct wave from the source and the reflected wave. The microphone will then receive both the direct wave from the source, as well as the reflected wave. This can be written mathematically as:

$$p_{tot}(t) = p_{dir}(t) + p_{ref}(t) \quad (10)$$

Consequentially, this relation leads to interference between waves. Interference describes the resulting effect of superposing two or more waves, where the resulting amplitude depends on their relative phase and amplitudes [3]. Generally, it is necessary to differentiate between two types of interference:

- Constructive interference – the waves arrive with similar phases, resulting in increased magnitude.
- Destructive interference – the waves arrive with opposite phases, resulting in reduced magnitude.

2.4.3 Path difference

The difference in paths implies a difference in travel distance. This is called *path difference*, and is described by:

$$\Delta r = r_{ref} - r_{dir} \quad (11)$$

For a wave of wavelength λ , the resulting phase difference due to the path difference can be expressed as [3]:

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta r \quad (12)$$

It is important to note that when microphones are placed at different positions along a line, the direct and reflected path lengths vary with microphone position. Consequently, the path difference and resulting phase difference also vary, leading to different interference patterns.

2.5 Diffraction & Insertion loss

When an acoustic wave encounters an obstacle, the phenomenon of *diffraction* may occur, in which the wave is “bent” around the obstacle. A direct consequence of this is that even when the direct path between a sound source and microphone is blocked, sound may still reach the microphone due to diffraction. The degree of diffraction depends on the wavelength relative to the characteristic size of the obstacle [3]. For a fixed obstacle, lower frequencies have longer wavelengths and therefore generally experience stronger diffraction, while higher frequencies are more strongly shadowed by the obstacle. The relationship

$$\lambda = \frac{c}{f} \quad (13)$$

shows that the wavelength decreases with increasing frequency.

Insertion loss (IL) describes the reduction in sound pressure level at a receiver position caused by the introduction of a noise barrier. It is defined as the difference between the sound pressure level without and with the barrier [4]:

$$IL = SPL_{no \text{ screen}} - SPL_{with \text{ screen}}. \quad (14)$$

Thus, a positive insertion loss indicates that the screen reduces the sound pressure level at the receiver position. In this work, insertion loss is calculated separately for each microphone position and frequency.

As stated above, the degree of diffraction is dependent on the wavelength in comparison to the size of the obstacle. Considering Equation 13, we therefore expect the insertion loss to depend on the frequency, as the wavelength changes relative to the dimensions of the obstacle [4].

3 Methodology

Based on the TLM method, as well as other principles explained in section 2, three different experiments were carried out. The experiments were:

- 1 – Free field propagation
- 2 – Reflection from a surface
- 3 – Noise screen

The methodology needed to conduct each experiment is described below.

3.1 TLMfig simulation setup

In order to carry out the experiments, the software TLMfig was used. TLMfig is a MATLAB-based graphical user interface for TLM simulations, where acoustic wave propagation is simulated numerically in 2D. The TLMfig source code was provided as part of the work, while the MATLAB commands used to configure the simulations and extract the microphone data were developed for the experiments in this work. The appendices cover code-related matters.

For all experiments, a grid of 200×200 cells was used, and a cell size of 0.1×0.1 m. This corresponds to a simulation area of 20×20 m. The grid was configured in the code by setting:

```
1 data.tlm_res = 200;  
2 delta_x = 0.1; % cell size [m]
```

For each experiment, a source was placed on the grid in the simulation, as well as microphone(s). The exact configuration is covered respectively in the description of each experiment later. Extraction of microphone data from TLMfig was done using the following code:

```
1 data = guidata(gcf);  
2 p = data.pmic;
```

3.2 Experiment 1 – Free field propagation

The simulation was set up with a sinusoidal source and an array of microphones along a common line extending from the source. The setup consisted of:

- A sine source at $[42, 89]$ with 15 cells/wavelength (NCPW).
- An array of microphones at coordinates $[56, 89], [72, 89], [88, 89], [104, 89], [121, 89]$.

The simulation was run for a total of 1000 time steps. The pressure data from each microphone were extracted and used to calculate the sound pressure level as a function of distance from the source. To exclude transient periods, the waterfall plot was examined to determine an appropriate time interval for the analysis.

3.3 Experiment 2 – Reflection from a surface

The simulation consisted of two setups with identical geometries, yet distinct reflection factors R ; specifically $R = 1$ for a hard surface and $R = 0.5$ for a partially absorbing surface. A Gaussian source was placed on a common line with five microphones, while a surface was placed below and parallel to this line. Apart from the different value of R , both setups consisted of:

- A Gaussian source at $[53, 82]$ with 45 cells/wavelength (NCPW)
- An array of 5 microphones evenly spaced on a common line with the source.
- A surface at $y = 20$

The simulation was run for a total of 500 time steps. The pressure data from each microphone were extracted and used to compare the resulting sound pressure levels for the two reflection factors.

3.4 Experiment 3 – Noise screen

The simulation was set up with a noise source positioned 0.5m above a flat surface with reflection factor $R = 0.95$. A roughly 3m high noise screen was placed 4m away from the source. An array of five microphones was positioned on the opposite side of the screen, 2.3m above the ground, at increasing distances from the screen, ranging from 0.4 to 8.9 m from the screen. The setup was simulated for both low- and high-frequency noise, corresponding to 15 and 45 cells per wavelength (NCPW), respectively. Additionally, in order to calculate insertion loss, simulation was conducted both with and without the noise screen present in the field.

The specifications of the simulation were:

- A noise source at coordinates $[12, 25]$ with respectively 15 and 45 NCPW
- A surface with reflection factor $R = 0.95$ at $y = 20$
- A noise screen ($R = 1$) at $x = 52$
- An array of five microphones at coordinates $[56, 43]$, $[77, 43]$, $[98, 43]$, $[119, 43]$, $[141, 43]$

For each frequency, simulations were performed both with and without the screen in order to calculate the insertion loss. The simulations were run for a total of 500 time steps. The pressure data from each microphone were extracted and used to calculate the insertion loss as a function of microphone distance.

3.5 Data processing

For each microphone, the extracted pressure signal was processed to obtain the RMS pressure and subsequently converted to sound pressure level using Equation 3.

For Experiment 1, the sound pressure level was calculated for each microphone position. The resulting SPL values were compared as a function of distance from the source to investigate the attenuation of sound pressure level during free-field propagation. Transient periods were excluded from the analysis by examining the waterfall plot and selecting an appropriate time interval.

For Experiment 2, the sound pressure level was calculated separately for each microphone and for both reflection factors, $R = 1$ and $R = 0.5$. The resulting values were compared between the two surface conditions to investigate the effect of the reflected sound field.

For Experiment 3, the sound pressure level was calculated for each microphone position for both the simulations with and without the noise screen. The insertion loss was then calculated as

$$IL = SPL_{\text{no screen}} - SPL_{\text{with screen}}. \quad (15)$$

This calculation was performed separately for the low- and high-frequency simulations, corresponding to 15 and 45 cells per wavelength, respectively.

4 Results

Following the methodology described in the previous section, data were obtained from each experiment. Note that the complete scripts used to process data for each experiment can be found in Appendix B.

4.1 Results of experiment 1

Pressure data from each of the five microphones was extracted from TLMfig, and used to calculate RMS pressure according to Equation 2. The corresponding relative sound pressure level was then calculated using Equation 3, with the sound pressure level at the first microphone used as the reference.

The resulting relative sound pressure levels are shown in Figure 2. Figure 3 shows a comparison between the experimental measurements and the theoretical expectation for two-dimensional free-field propagation, as described in Equation 5.

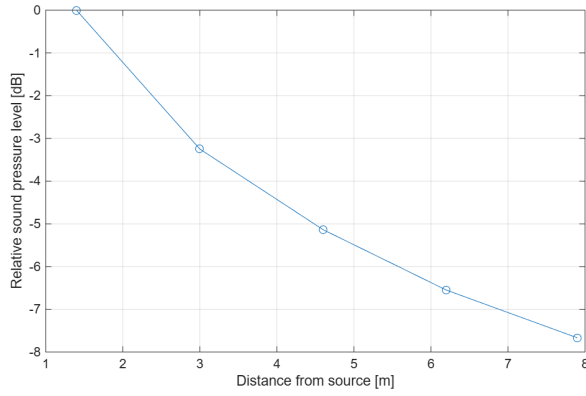


Figure 2: SPL in experiment 1 relative to microphone 1.

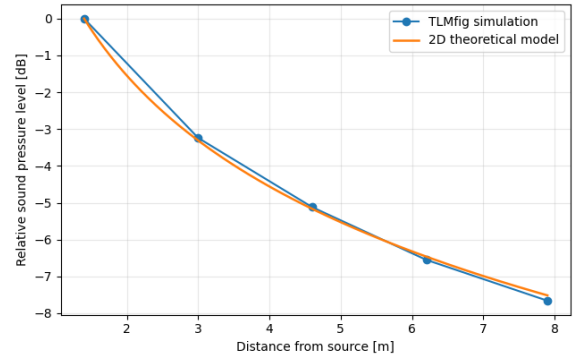


Figure 3: Comparison between the experimental measurement and the theoretical expectations.

As seen in both figures, the SPL decreases with increasing distance from the source. The first microphone was located 1.4m from the source and was used as the reference.).

The relative SPL decreases from 0 dB at the reference microphone to approximately -7.7 dB at the furthest microphone. The experimental measurements follow the general trend predicted by the theoretical free-field relationship, although slight deviations are observed at some microphone positions.

4.2 Results of experiment 2

Pressure data from each of the five microphones was extracted from TLMfig, and used to plot pressure change as a function of time. Figure 4 and Figure 5 show respectively the measurements for a simulated hard and partly absorbing ground. The first cluster in each plot corresponds to the arrival of the direct wave, while the later cluster corresponds to the wave reflected from the ground surface.

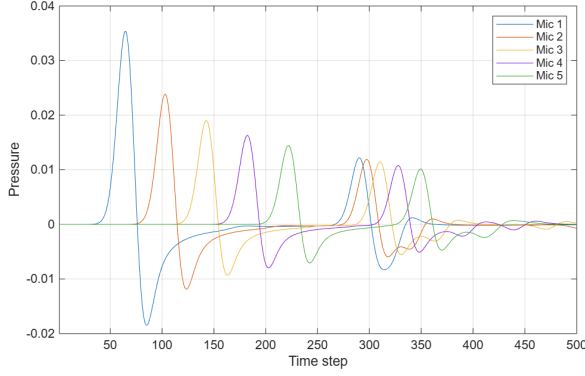


Figure 4: Pressure measurements at the five microphones for a hard ground with reflection factor $R = 1.0$.

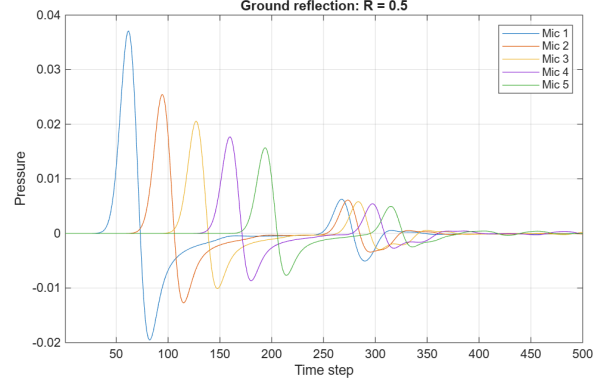


Figure 5: Pressure measurements at the five microphones for a partly absorbing ground with reflection factor $R = 0.5$.

The second simulation had to be configured independently because the field could not be edited after completion of the first simulation. Consequently, small differences in source and microphone placement may have occurred between the two simulations. This limitation is discussed further in section 5.

Besides the inaccuracies in source and microphone placement, the direct wave appears to be similar in both simulations. The reflected wave, however, shows a noticeable reduction in amplitude for the partly absorbing surface. For the hard surface ($R = 1$), the reflected wave reaches a peak pressure of approximately 0.12 Pa, while the corresponding peak for the partly absorbing surface ($R = 0.5$) is approximately 0.06 Pa. This corresponds to a reduction in reflected peak pressure of approximately 50 %.

4.3 Results of experiment 3

Pressure data from each of the five microphones was extracted from TLMfig, and used to calculate insertion loss due to the noise screen. The calculations were carried out for both low frequencies (NCPW of 15) and high frequencies (NCPW of 45), and were based on simulation both with and without the noise screen. Figure 6 shows the insertion loss as a function of distance from the screen.

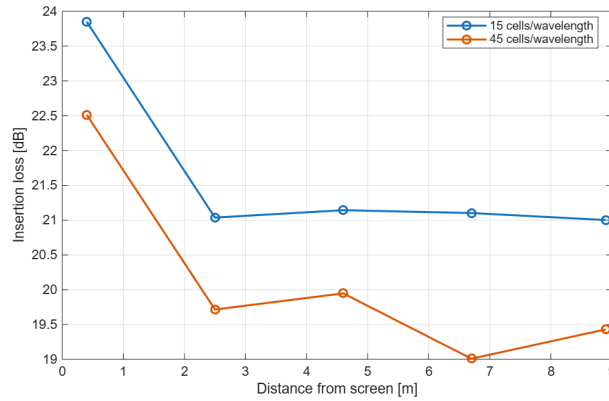


Figure 6: Insertion loss as a function of distance from noise screen for different frequencies.

It should be noted that the noise screen had to be manually drawn for both frequency simulations, which may have introduced small differences in its position between the simulations. This is further discussed in section 5.

The insertion loss varies with microphone distance for both frequencies. For 15 NCPW, the insertion loss ranges from approximately 21.0 to 23.9 dB, while for 45 NCPW it ranges from approximately 19.0 to 22.5 dB. The insertion loss is generally higher for the 15 NCPW case, corresponding to the higher-frequency simulation. This is consistent with the expected reduction in diffraction at higher frequencies.

5 Discussion

In this section, the results of the experiments will be reviewed and discussed. The impact of error-sources will be examined, and suggestions for improvements will be provided.

5.1 Comments on experiment 1

The intention of experiment 1 was to utilize TLM simulation in order to examine the variation in sound pressure level (SPL) during free field propagation. Figure 3 shows a clear resemblance between the simulated SPL and the theoretical relationship for two-dimensional free-field propagation described by Equation 5. The results show a reduction in SPL with increasing distance, consistent with the theoretical expectations provided through Equation 8.

It should be noted that the results were expressed relative to the SPL at the first microphone, which was located 1.4, m from the source. This does not affect the expected distance dependence, but a reference position closer to the source would have provided a larger range over which to compare the simulated and theoretical relationships.

5.2 Comments on experiment 2

Experiment 2 was designed to examine the effect of introducing a reflective ground surface in the field. In Figure 4 and Figure 5, the reflected waves can clearly be distinguished from the direct waves by their later arrival time, at approximately time step 300. This is consistent with the longer propagation path of the reflected wave, as described by the path difference in Equation 11. The longer propagation path also contributes to the lower amplitude of the reflected wave, consistent with the distance dependence observed in Experiment 1.

For $R = 0.5$, the magnitude of the reflected wave is approximately half that observed for $R = 1$. This is consistent with the specified reflection factor. The direct wave remains approximately unchanged between the two simulations, as expected since the ground surface should only affect the reflected propagation path. Exact numerical estimation of the reflection factor from the simulated signals is beyond the scope of this experiment.

As discussed in section 4, the experiment was subject to potential errors due to limitations in the control of the simulation setup. Small differences in the source and microphone positions may therefore have affected the comparison between the two simulations. Within the scope of the experiment, however, the results are consistent with the expected effect of changing the reflection factor.

5.3 Comments on experiment 3

The set up in experiment 3 was intended to examine how well an acoustic screen reduced noise from a source. Insertion loss was used as a measure of the reduction in sound pressure levels resulting from introducing the noise screen. As shown in Figure 6, the screen produced a substantial reduction in SPL at the microphone positions.

The insertion loss was generally greater for the higher-frequency noise (15 NCPW) than for the lower-frequency noise (45 NCPW). This is consistent with the theoretical discussion of diffraction in section 2. For a fixed screen geometry, the higher-frequency signal has a shorter wavelength and is therefore less strongly diffracted around the screen. Consequently, the screen provides greater attenuation at the higher frequency.

It is necessary to point out that the noise screen had to be redrawn manually between the simulations. Small differences in its position may therefore have been introduced. In addition, the intended microphone height was approximately 1.5, m above the ground, whereas the resulting microphone coordinates corresponded to a height of approximately 2.3, m. This deviation resulted from the limitations of manually

positioning elements in TLMfig. Both effects introduce potential sources of uncertainty in the comparison between the simulations. More precise control of the source, screen, and microphone positions would improve the reproducibility of the experiment.

6 Conclusion

The intention of this work was to examine the use of TLM as a numerical simulation method for acoustic wave propagation. Three experiments were conducted, investigating free-field propagation, reflection from ground surfaces with different reflection factors, and noise reduction using an acoustic screen.

The results were generally consistent with the theoretical expectations. In the free-field experiment, the sound pressure level decreased with increasing distance in accordance with the expected two-dimensional propagation relationship. The reflection experiment demonstrated the effect of the ground reflection factor, with a lower reflection factor resulting in a reduced reflected wave amplitude. The noise-screen experiment showed that the screen produced a significant reduction in sound pressure level, with greater insertion loss observed for the higher-frequency case (15 NCPW).

Overall, TLMfig was found to be a useful tool for illustrating fundamental acoustic propagation phenomena. However, the simulations were subject to some non-negligible sources of uncertainty, particularly due to limitations in the simulation interface and the manual placement of sources, microphones, and the noise screen. Despite these limitations, the simulations produced results that were broadly consistent with the theoretical expectations.

References

- [1] Y. Kagawa, T. Tsuchiya, B. Fujii, and K. Fujioka, “Discrete huygens’ model approach to sound wave propagation,” *Journal of Sound and Vibration*, vol. 218, no. 3, pp. 419–444, 1998.
- [2] A. Bjerkvik, *TLMfig User’s Guide*, NTNU, n.d.
- [3] S. L. Garrett, *Understanding Acoustics: An Experimentalist’s View of Sound and Vibration*, 2nd ed. Springer, 2020.
- [4] Federal Highway Administration, “Noise barrier design handbook: 3. acoustical considerations,” https://www.fhwa.dot.gov/ENVIRonment/noise/noise_barriers/design_construction/design/design03.cfm, U.S. Department of Transportation, Federal Highway Administration.

A Appendix A - Declaration of AI use

This appendix consists of only of a brief statement regarding the use of *artificial intelligence* (AI) during both the conduction of the experiment, as well as the making of the report.

All use of AI have been conducted in accordance with the Acta Acustica [ref] guidelines, i.e. that AI is used as a *tool*, rather than as a source of information nor to generate contents in the experiment/report. When used, the AI in question would be ChatGPT (pirmarily model 5-6). The aid of AI was applied to *exclusively* the following aspects of the process;

- In the report itself, as feedback on grammar and contents. Note that there is, under no circumstances, integrated AI-generated text in the report. All text-based contents are written by the author, with AI only offering grammatical through review of contents.
- In the code segments, AI was used to write code efficiently. None of the generated code was implemented without a thorough manual check, in which the author made sure that the code a) worked, and b) was in conformity with the theory and method described in the report. In other words, AI was used only as a tool in order to speed up the process of writing already familiar code (such as simple display of figures).

B Appendix B – MATLAB scripts

The MATLAB scripts used for the simulations, extraction of microphone data, data processing, and plotting are provided as separate files in the submitted .zip folder. The folder contains scripts for all three experiments, as well as provided code for running TLMfig.

The scripts include code for extracting microphone pressure data from TLMfig, calculating RMS pressure and sound pressure level, calculating insertion loss, and generating the plots presented in the report.

The .zip file might also be retrieved from <https://github.com/Eek-JensenCEO/TTT4181-Assignment-1.git>

C Appendix C - Comments on TLMfig curiosities

This appendix will simply inform of bugs and/or strange behaviours from the TLMfig software during the experiments.

Problem regarding number of microphones in array

When drawing an array of microphones, the last one would only appear in the GUI, not when trying to extract position data. This was fixed by changing the following line of code:

```
1 data.geometri(floor(y2),floor(x2))=data.geometri(floor(y2),floor(x2))+10;
```

To

```
1 data.geometri(floor(y2),floor(x2))=data.geometri(floor(y2),floor(x2))+100;
```

I.e. simply a change from +10 to +100. This fix was suggested by ChatGpt, and was kept throughout all experiments.

Problem with editing after ended simulation

In experiment 2 and 3, running subsequent simulations with consistent, geometrical properties was expected. As discussed in section 5, after the simulation was completed, the only way to alter the field was to restart the field. Thus, every element placed in the field would need to be replaced, and, apart from the lines, the author found no way to consistently place elements at exact coordinates. This would unavoidably lead to unintended structural inaccuracies in the field between simulations.

It should be noted that this problem could be (partially) due to user inexperience, as the author was able to edit the simulation for experiment 3. Yet it remains clear that the UI needs an overhaul.